NeoPixels

DO | PICO | OUTPUT | LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses



Gaming PC lighting

DETECT Read lighting selection → **DECIDE** Which colour should each pixel be? → **DO** Light individual pixels



Disco shoelace meter

DETECT Read shoe movement → **DECIDE** Which light pattern should run? → **DO** Animate the pixels



Pixel RGB LED and controller

IN Data from the board

OUT Data to another pixel

VDD and GND 3.3 V power

Raspberry Pi Foundation

Code

Raspberry Pi Pico

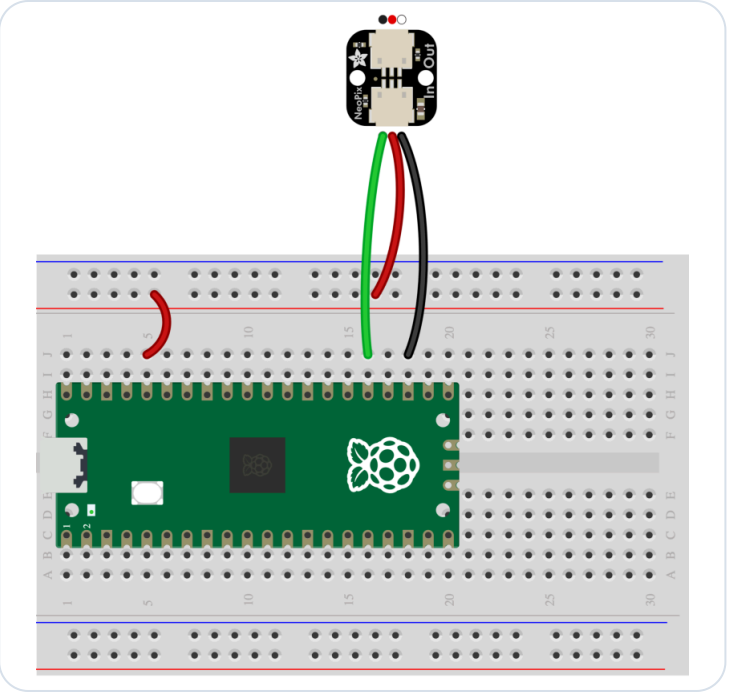
```
from machine import Pin
from neopixel import NeoPixel

pixels = NeoPixel(Pin(19), 1)
pixels[0] = (255, 0, 80)
pixels.write()
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

Wiring

PICO



GP19 (pin 25) → IN, **3V3 OUT** (pin 36) → top + rail → VDD, **GND** (pin 23) → GND